

	deduce whether a beta particle is due to the decay of strontium-90 or yttrium-90.	
7ai	1. Solar energy can be affected by cloud cover, OR is only present in the day. 2. Wind speeds can vary. 3. It is difficult to efficiently store excess energy generated for use at a later timing when the supply of energy falls. (any 2 of the above)	B1 B1
7aii	γ -radiation is <u>high energy</u> electromagnetic radiation OR photons emitted from radioactive decay of atomic nuclei.	B1 B1
7aiii	γ -radiation can penetrate the skin while its ionizing powers can cause damage to DNA / cells / organs.	B1 B1
7aiv	To ensure that all γ -radiation would travel the same distance x through the absorber.	A1
7av	The curve reaches an asymptote at the x -axis.	A1

7bi	$C_x / C_0 = e^{-\mu x}$, Taking ln on both sides, $\ln (C_x / C_0) = -\mu x$ As Fig. 7.3 is a graph of $\ln (C_x / C_0)$ against x with a <u>straight line, negative gradient, passing through the origin,</u> it indicates a relationship $C_x / C_0 = e^{-\mu x}$.	B1 B1 B1
7bii	gradient = $-\mu$ gradient = $-4.5 / 10 = -0.45$ Hence, $\mu = 0.45 \text{ cm}^{-1}$	M1 A1
7ci	units of $\mu_m = \text{units of } \mu / \text{units of } \rho$ $= \text{cm}^{-1} / \text{g cm}^{-3}$ $= \text{g}^{-1} \text{ cm}^2$	A1
7cii	For lead, $\mu = 0.45 \text{ cm}^{-1}$ $\mu_m = \mu / \rho = 0.45 / 11.3 = 0.0398 = 0.040 \text{ cm}^{-1}$	A1
7di	average $\mu_m = (0.035 + 0.037 + 0.040) / 3 = 0.037 \text{ cm}^2 \text{ g}^{-1}$	A1
7dii	For concrete, $\mu = \mu_m \rho = 0.037 \times 2.4$ $= 0.037 \times 2.4 \times 10^3 \times 10^3 / 100^3$ $= 0.0888$ $= 0.09 \text{ cm}^{-1}$	M1 M1 A0
7diii	$C_x / C_0 = e^{-\mu x}$, For same shielding effect, value of C_x / C_0 is the same. Hence, value of μx must be the same. $(\mu x)_{\text{concrete}} = (\mu x)_{\text{lead}}$ $(0.09) x = (0.45) (4.0)$ $x = 20 \text{ cm}$ OR From Fig. 6.2, when $x = 4.0 \text{ cm}$, $C_x / C_0 = 0.16$ Using $C_x / C_0 = e^{-\mu x}$, $\ln (C_x / C_0) = -\mu x$ $\ln 0.16 = -0.09 x$ $x = 20 \text{ cm}$	C1 A1 C1 A1
7div	1. Concrete is cheaper OR more available than lead. 2. Concrete is a stronger material than lead OR Concrete is a better choice as a construction material than lead. 3. Lead is toxic compared to concrete.	B1 B1